DASC 3240, L06

Plotting one variable – Assignments

2023/24 SPRING SEMESTER

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Today's outline

- Plotting one variable – Assignments
  - Data
  - Goals of today's assignments

Data

Datasets

Rdatasets 1.0.0 Available datasets

What is this?

`Rdatasets` is a collection of 2264 datasets which were originally distributed alongside the statistical software environment `R` and some of its add-on packages. The goal is to make these data more broadly accessible for teaching and statistical software development.

What is included?

The list of available datasets (csv and docs) is available here:

- HTML index
- CSV index

On the github repository you will also find the scripts I use to scrape data and update the website.

https://vincentarelbundock.github.io/Rdatasets/

ChickWeight

https://vincentarelbundock.github.io/Rdatasets/doc/datasets/ChickWeight.html
https://vincentarelbundock.github.io/Rdatasets/doc/datasets/ChickWeight.csv

Weight versus age of chicks on different diets

Description

The ChickWeight data frame has 578 rows and 4 columns from an experiment on the effect of diet on early growth of chicks.

	weight	Time	Chick	Diet
1	42	0	1	1
2	51	2	1	1
3	59	4	1	1
4	64	6	1	1
5	76	8	1	1
6	93	10	1	1

weight the body weight of the chick (gm).
Time the number of days since birth when the measurement was made.
Chick levels 18 < ... < 48 giving a unique identifier for the chick. The ordering of the levels groups chicks on the same diet together and orders them according to their final weight (lightest to heaviest) within diet.
Diet levels 1, ..., 4 indicating which experimental diet the chick received.

chickwts

https://vincentarelbundock.github.io/Rdatasets/doc/datasets/chickwts.html
https://vincentarelbundock.github.io/Rdatasets/doc/datasets/chickwts.csv

Chicken Weights by Feed Type

Description

An experiment was conducted to measure and compare the effectiveness of various feed supplements on the growth rate of chickens.

```
  weight      feed
1    179 horsebean
2    160 horsebean
3    136 horsebean
4    227 horsebean
5    217 horsebean
6    168 horsebean
```

weight the chick weight.
feed the feed type.

InsectSprays

https://vincentarelbundock.github.io/Rdatasets/doc/datasets/InsectSprays.html
https://vincentarelbundock.github.io/Rdatasets/csv/datasets/InsectSprays.csv

Effectiveness of Insect Sprays

Description

The counts of insects in agricultural experimental units treated with different insecticides.

```
  count spray
1    10     A
2     7     A
3    20     A
4    14     A
5    14     A
6    12     A
```

[,1]	Insect count
[,2]	The type of spray

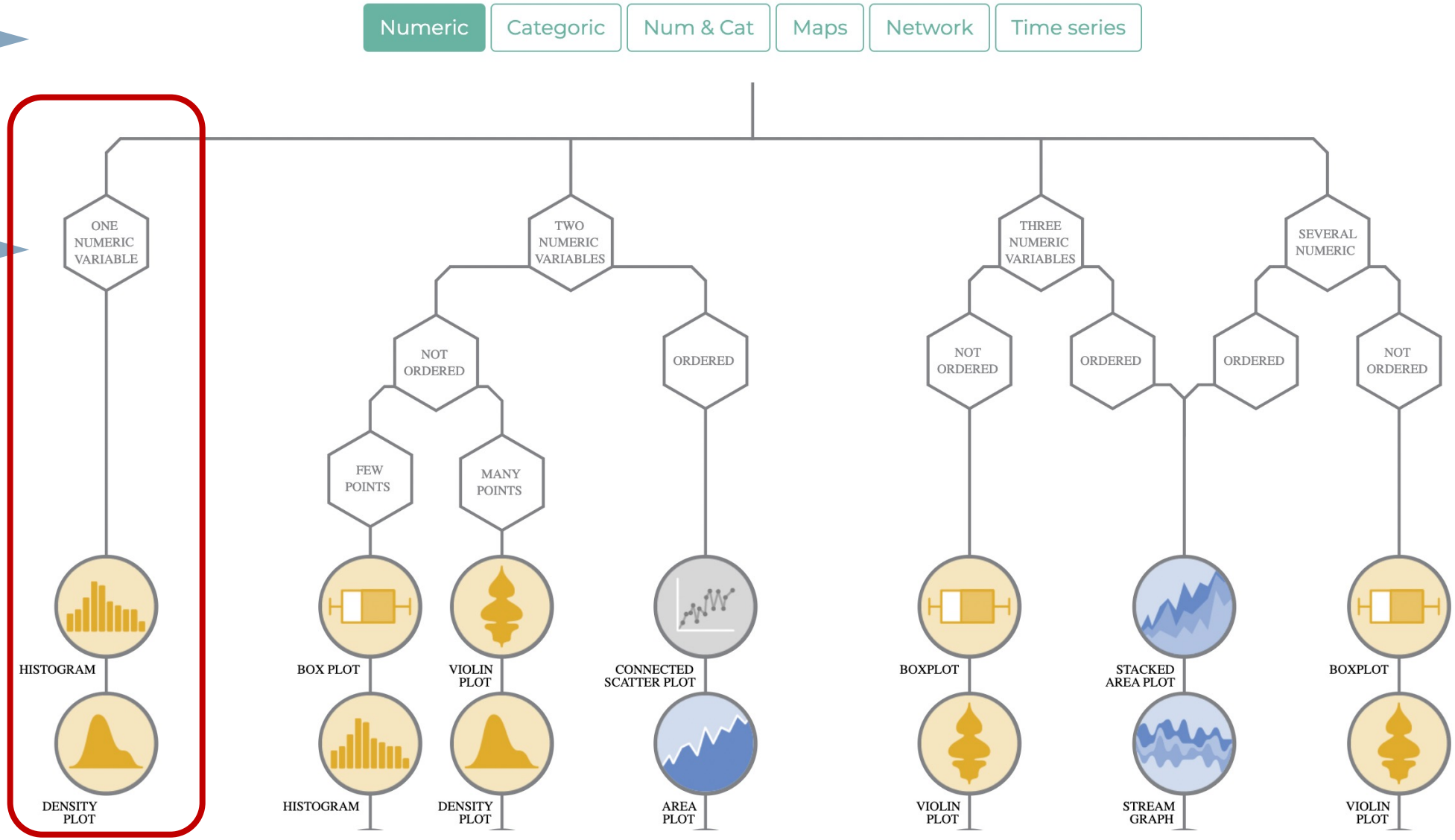
Goals of today’s assignments

HISTOGRAM AND DENSITY PLOT

How many variables? Numeric or categorical?

Numeric or categorical?

The number of variables



https://www.data-to-viz.com/

Assignments

- **ChickWeight**
  1. Create a simple histogram of weight. Please use all data.
  2. Exclude data with Time < 10 and create histogram of weight for each “Diet”. There are four Diet, so you will have four histograms, but please overlay all the histograms in one figure.
  3. Draw each histogram in each separate panel (facet) and combine them in a 2 x 2 format. You will have four facets in one large figure.

- Please **add the sufficient information (e.g., figure titles, axis labels, legends, and so on) in the figures** so that readers can understand which histogram indicates which category!
- Please **be careful with the colors, fonts, and layout** so that your figures are beautiful and meaningful to people with different backgrounds.
- During the class, please discuss how you can effectively visualize the effect of treatments. Please implement better visualizations and share the idea with your colleagues.

Assignments

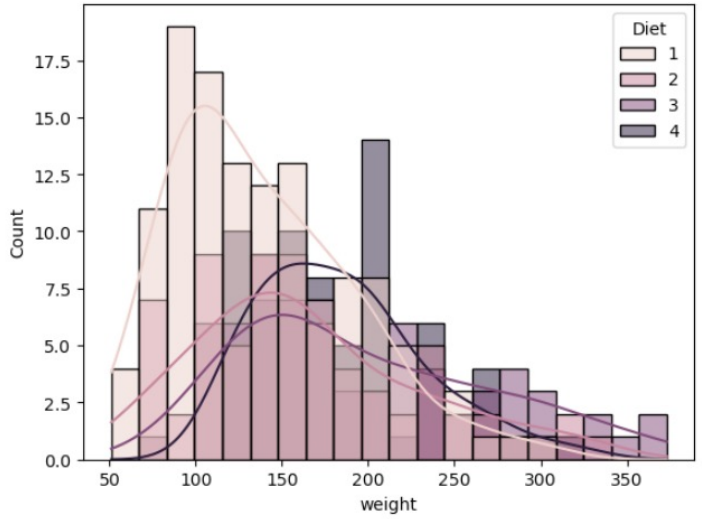
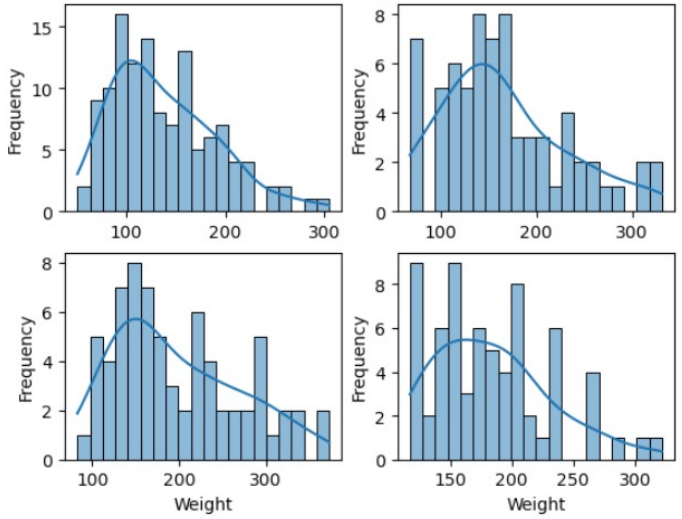
Optional assignments

- **chickwts**
  1. Create histogram of weight for each “feed”. Please combine the four histograms into one figure.
- **InsectSprays**
  1. Create histogram of count for each “spray”. Please combine the four histograms into one figure.

Assignments: Example figures

- **ChickWeight**

2. Exclude data with Time < 10 and create histogram of weight for each “Diet”. There are four Diet, so you will have four histograms, but please overlay all the histograms in one figure.



3. Draw each histogram in each separate panel (facet) and combine them in a 2 x 2 format. You will have four facets in one large figure.